KaxaNuk

Sharing knowledge

Backtesting & Attribution Analysis

Session 4

Content

- The Simulation Challenge in Investment Research
- Understand Your Benchmark
- Backtesting
- Factor Models Insights
- Attribution Analysis

Where We Are in the Research Process

In Session 1 we discussed: The Evolution of Investment Research

Observation → Theory → Models → Systems

In Session 2 we discussed: Data Curation and Feature Engineering

Raw Data → Structured Data → Features → Signals

In Session 3 we discussed: Portfolio Construction and Strategy Modeling

Signals → Selection → Sizing → Timing → Historical Portfolios

Now we answer the question:

How do we know if a strategy actually works?

The Simulation Challenge in Investment Research

Bottlenecks

The Data Challenge

60–80%

of a researcher's time may be wasted on repetitive data cleaning and structuring tasks.

Reproducibility Crisis

- Undocumented workflows make results impossible to validate or replicate.

Scattered Sources

- Inconsistent provider formats require constant, error-prone normalization.

Manual Errors

- Human intervention introduces silent data failures that are hard to detect and harder to fix.

Lost Velocity

- Every hour spent fixing pipelines is an hour not spent generating alpha.

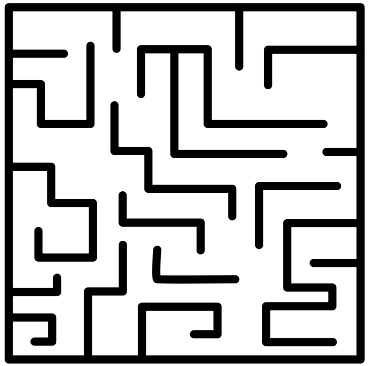
The Strategy Challenge

Finding a predictive signal is only the beginning.

The real challenge is building a complete strategy system.

Researchers must answer several structural questions:

- What securities can we invest in?
- How do we construct portfolios?
- How do we simulate historical performance?
- How do we understand what drives results?



Investment research is not just about signals.

It is about designing systems that allocate capital.

The Simulation Challenge

Historical portfolios allow us to simulate. But simulation requires discipline.

In Session 3 we built historical portfolios by applying selection, sizing, and timing rules through time.

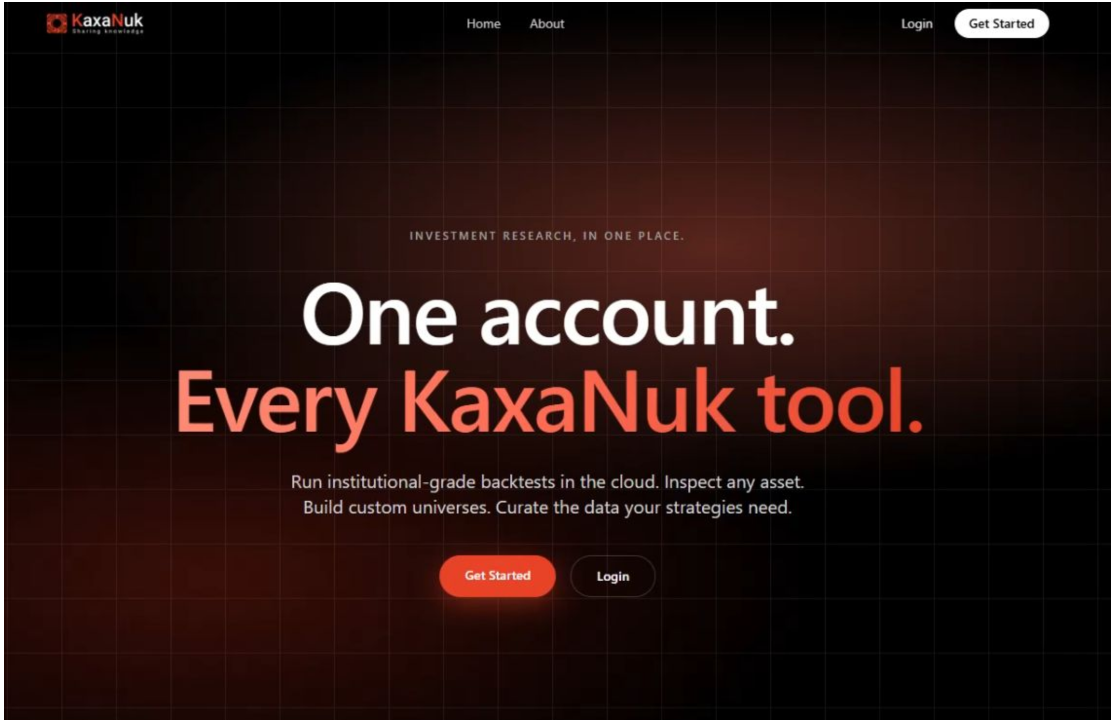
The result was a time series of portfolio weights and returns.

Now the question changes:

Is that time series telling us the truth?

A simulation is only as credible as the data and process behind it. Every shortcut taken in construction becomes a source of error in evaluation.

One Account. Every KaxaNuk Tool.



Understand Your Benchmark

Like-For-Like

A Strategy Without a Benchmark Is Directionless

Capital is always allocated relative to something. Make that something rigorous.

A benchmark defines:

➔ The eligible investable universe
➔ What counts as outperformance
➔ The baseline risk exposures your strategy must beat
➔ The governance and reporting standard for your results

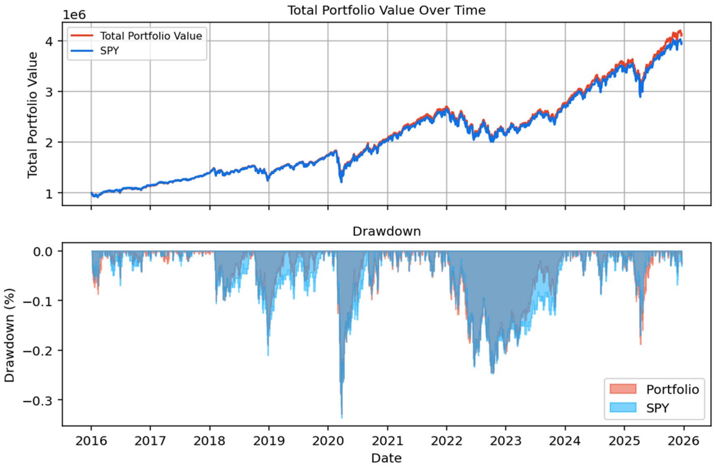
Without a rigorous benchmark:

➔ You cannot separate alpha from beta
➔ Sector tilts look like skill
➔ Relative performance evaluation becomes meaningless

The KN US Equity Benchmark

Our Reference Point for This Challenge

- ➔ **Universe: Top 450 US companies by market capitalization**
- ➔ Period: January 2015 – Today
- ➔ Construction: Rule-based, transparent, fully documented
- ➔ **Provides: Eligible investable universe · Daily historical weights · Full historical return series**



Backtesting

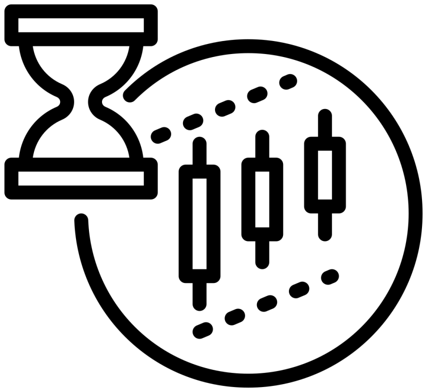
Back To The Future

A Backtest Is Only as Good as Its Assumptions

Backtests look precise. They rarely are.

They quietly depend on:

- Point-in-time data integrity
- Look-ahead bias controls
- Survivorship bias handling
- Realistic rebalancing logic
- Execution price assumptions
- Transaction cost modeling
- Capital path dependency



Small implementation differences can radically alter results.

Backtesting should be controlled experimentation, **not optimistic simulation**.

The Hidden Pitfalls

Know what can silently destroy your backtest

Look-Ahead Bias

- ➔ Using data not available at decision time. Example: rebalancing using earnings reported two weeks after the period closes. In Session 2, we built point-in-time pipelines precisely to prevent this.

Survivorship Bias

- ➔ Testing only on companies that survived. Bankrupt and delisted companies are excluded from the universe — performance is artificially inflated.

Overfitting

- ➔ Parameters tuned so tightly to history that the strategy captures noise, not signal. The SMA(50)/SMA(200) crossover from Session 3 was chosen for its simplicity and documented academic support — not because it backtested best.

Data Snooping

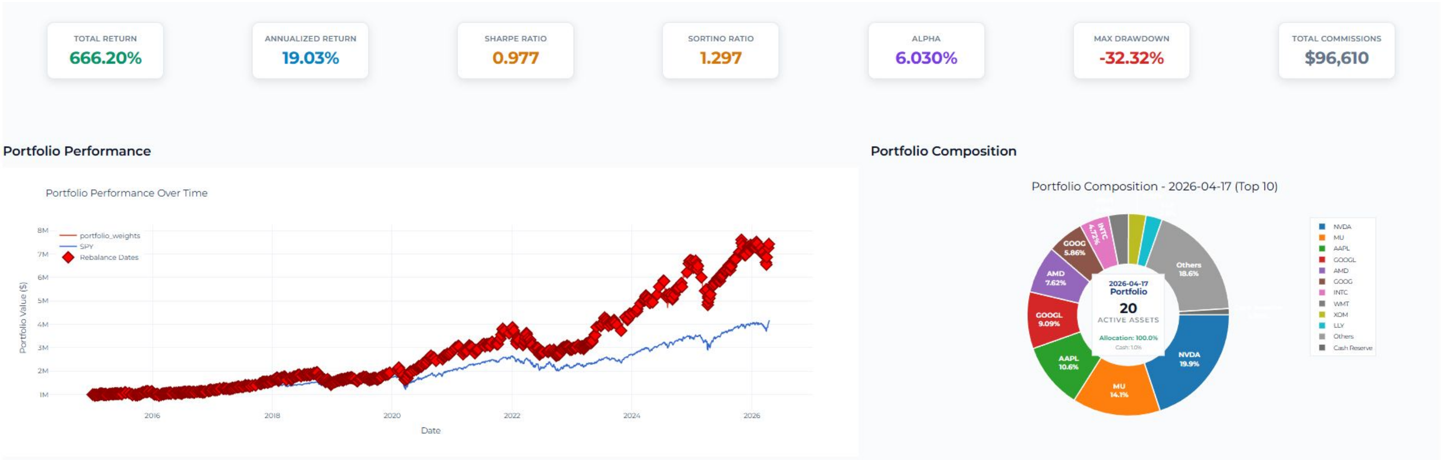
- ➔ Testing hundreds of variations until one passes. The more combinations you try, the more likely a good result is accidental.

Key Backtest Metrics

How to read your results

Performance metrics alone are not sufficient to validate a strategy.

They must be paired with robustness tests and attribution analysis.



Strategy Robustness

A good backtest is stress-tested, not just reported.

Robust strategies should be evaluated under different scenarios:

Out-of-sample performance

➔ Reserve a holdout period never touched during development. If the strategy only works in-sample, it is not a strategy.

Parameter sensitivity analysis

➔ Does performance collapse if SMA(50) becomes SMA(45) or SMA(55)? Robust strategies are stable across small parameter changes.

Market regime evaluation

➔ Does the strategy hold up in drawdowns, high volatility, and sideways markets — not just bull runs?

Diversification analysis

How concentrated is the portfolio? Is performance driven by one or two positions?

Factor Models

Every Return Has a Reason, Where is yours coming from?

The KN US Factor Model

Built to answer: where does performance actually come from?

With a the same universe used to calculate the KN US Equity Benchmark, and using the **Fama-McBeth** methodology to calculate the **factor exposure of all stocks to all factors**, instead of the standar **Fama-French** procedure, where they create a long short portfolio of the factor, ending with **the daily exposures of a portfolio**.

Using 5 Style Factors + 11 Industry Factors (GICS)

In Session 2, we discussed how features can be used as either alpha signals or risk factors.

The KN US Factor Model uses those same features, now structured to explain portfolio returns, not predict them.

Factor Description

- **Beta:** Returns sensitivity to our US500 index

- **Size:** Natural Logarithm of Market Capitalization

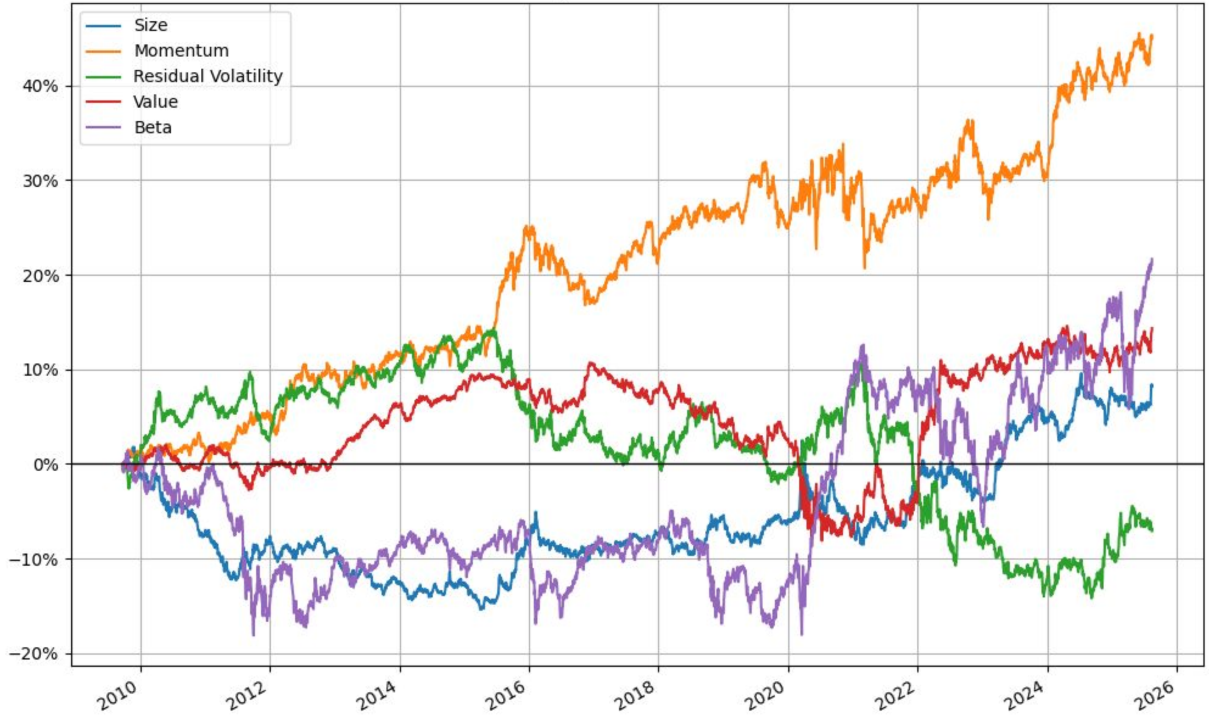
- **Value:** Average of Book-to-Price, Earnings-to-Price and Sales-to-Price

- **Residual Volatility:** CAPM regression standard deviation, Daily volatility and cumulative range indicator

- **Momentum:** CAPM alpha, relative strength indicator

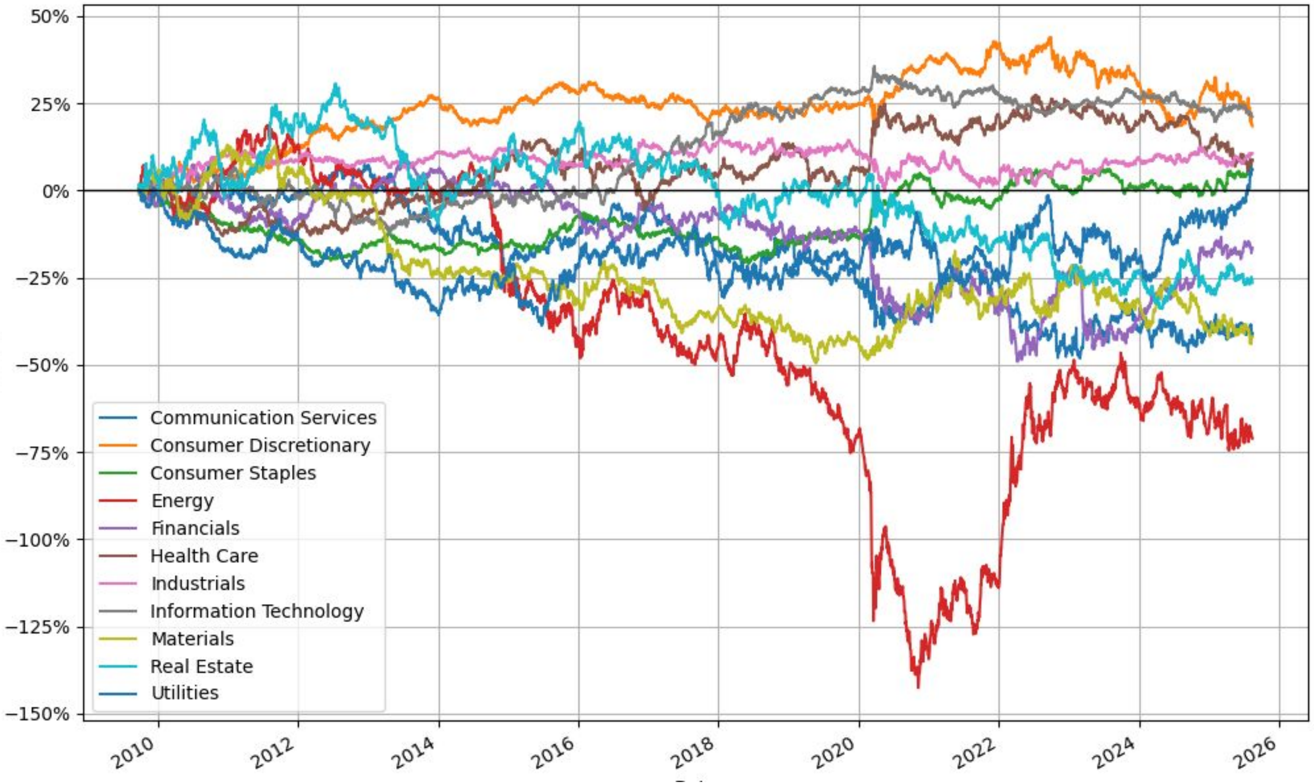
Our Style Factors

- Beta
- Size
- Value
- Residual Volatility
- Momentum
- Market



Our Industry Factors

The 11 GICs Sectors



Factor Passport Integrity

KaxaNuk Factor Passport

5 Style Factors + 11 Gics Industries + US Market

Oct 2009 to Aug 2025, 3992 Observations

Factor	Average Return	Annualized Vol	Info ratio	Cummulative Returns	Days above abs(2)	% days above abs(2)	Mean tstats	Mean vif
Communication Services	-3.16%	11.17%	-0.2827	-40.93%	872	21.84%	1.3327	1.0313
Consumer Discretionary	0.92%	7.05%	0.1302	18.41%	1,531	38.35%	1.8397	0.9415
Consumer Staples	0.39%	5.42%	0.0726	8.55%	1,012	25.35%	1.4123	1.0485
Energy	-5.81%	17.22%	-0.3374	-71.11%	1,918	48.05%	2.4625	1.0375
Financials	-1.54%	10.13%	-0.1518	-16.42%	1,632	40.88%	2.0519	1.0813
Health Care	0.25%	7.73%	0.0328	8.75%	1,716	42.99%	2.0679	0.9725
Industrials	0.52%	5.59%	0.0921	10.62%	1,306	32.72%	1.6885	0.9295
Information Technology	1.15%	6.20%	0.1849	21.12%	1,682	42.13%	2.0451	0.9712
Materials	-3.17%	10.71%	-0.2961	-41.95%	1,148	28.76%	1.5533	1.0083
Real Estate	-2.35%	11.88%	-0.1978	-26.46%	1,252	31.36%	1.6598	1.0585
Utilities	-0.28%	11.41%	-0.0245	5.88%	1,200	30.06%	1.6378	1.1195
Size	0.45%	3.76%	0.1187	8.17%	1,693	42.41%	2.0520	1.6862
Momentum	2.70%	5.90%	0.4572	44.93%	2,017	50.53%	2.5328	1.4257
Residual Vol	-0.59%	5.28%	-0.1111	-7.11%	1,972	49.40%	2.4728	1.3249
Value	0.83%	4.03%	0.2050	14.34%	1,373	34.39%	1.7688	1.1938
Beta	1.10%	7.37%	0.1494	21.65%	2,542	63.68%	3.7651	1.8924
Market	9.20%	17.75%	0.5186	164.60%	3,351	83.94%	10.4069	1.3761

Attribution Analysis

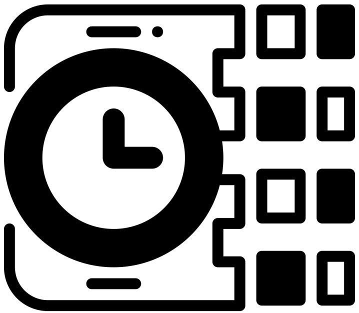
Know What You Own. Know Why It Worked.

Returns Must Be Decomposed

Performance Without Decomposition Is Incomplete.

Even strong results leave critical questions unanswered:

- Is it coming from exposure to market (beta)?
- From a specific sector tilt?
- From known risk factors?
- From unintended exposures?
- Or from true alpha?



Without structured attribution:

Conviction is misplaced. Risk is misunderstood. Capital allocation becomes fragile.

The Attribution Framework

Thanks to the Fama-McBeth methodology, we have a **factor exposure for each stock for each date (Factor Loadings)**, the exposure of the portfolio is just the average of all portfolio stock's weighted by their size.

So the return of the portfolio:

$$R_p = \beta_p R_m + \sum_{i=1}^{K} f_{p,i}\,\lambda_i + \sum_{j=1}^{S} s_{p,j}\,r_{s,j} + \alpha_p$$

In Matrix Form:

$$\mathbf{R} = \boldsymbol{\beta} R_m + \mathbf{f}\boldsymbol{\lambda} + \mathbf{S}\mathbf{r}_s + \boldsymbol{\alpha}$$

Multiply by portfolio weights:

$$\mathbf{w}^\top \mathbf{R} = \mathbf{w}^\top (\boldsymbol{\beta} R_m + \mathbf{f}\boldsymbol{\lambda} + \mathbf{S}\mathbf{r}_s + \boldsymbol{\alpha})$$

Understand Your Performance

Risk Model:

- What systematic exposures does your portfolio carry? Factor loadings quantify the strategy's structural tilts.

Attribution:

- How much return came from each exposure? Decompose performance into market, factor, sector, and selection components.

Alpha:

- What is left that cannot be explained by any known factor! True alpha is rare, small, and hard to sustain.

A good strategy has intentional exposures, documented tilts, and alpha that survives decomposition.

Understand Your Performance

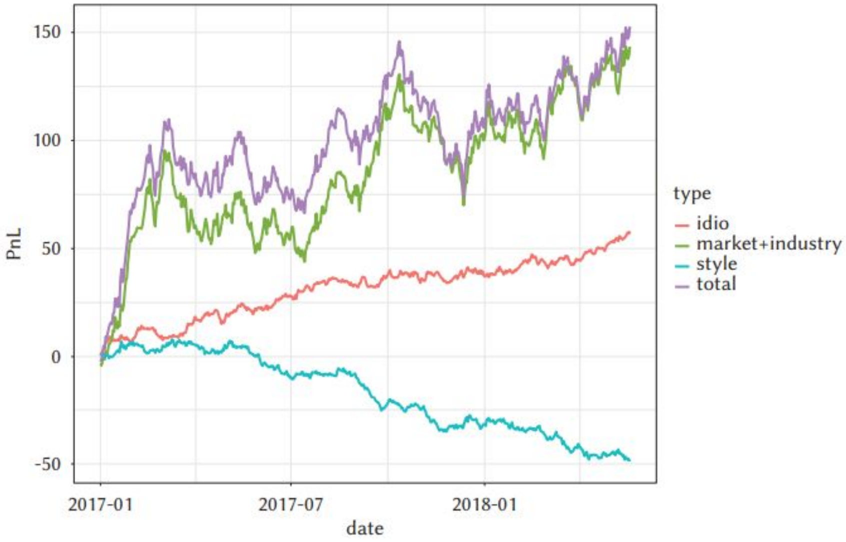


Figure 8.1 Performance Attribution example. The figure shows total PnL, idio PnL, market and industry PnL grouped together, and style PnL grouped together.

[PA20] Giuseppe Andrea Paleologo. Advanced Portfolio Management: A Quant's Guide for Fundamental Investors. 2020.

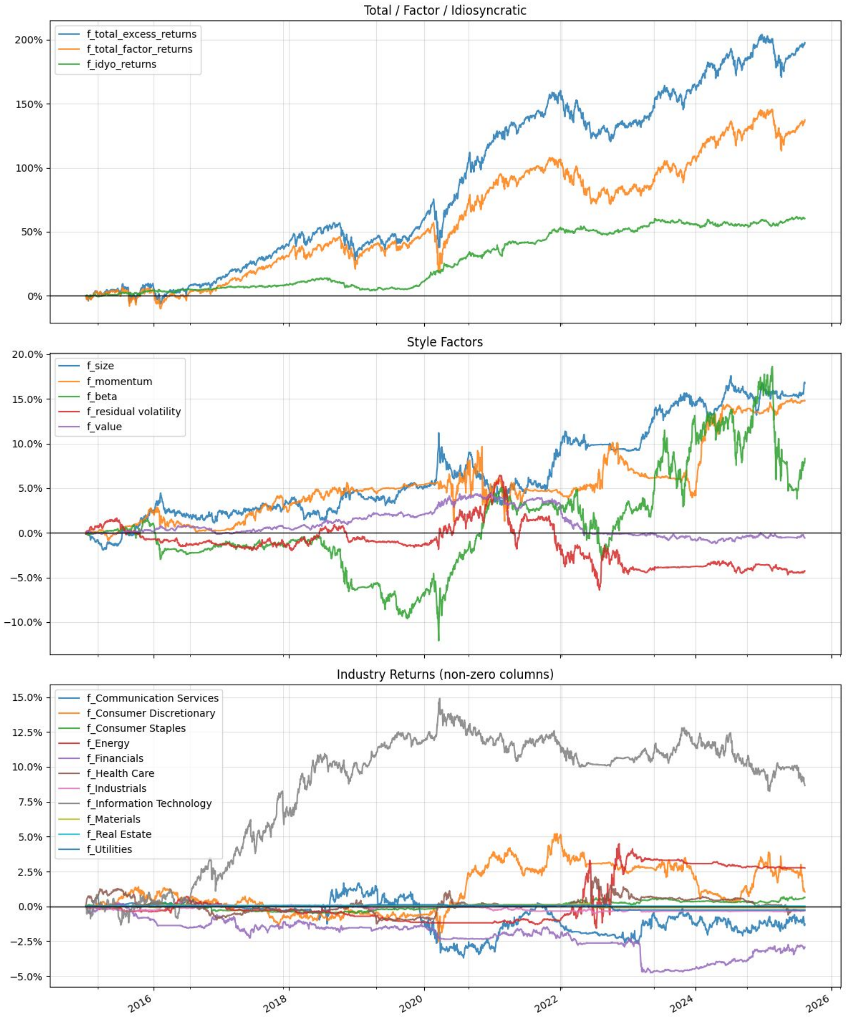
Attribution to the Strategy

Continuing on with the strategy from Session 3 we perform an Attribution Analysis with KN Factors

High momentum: crossover strat explicitly targets trend-following stocks

Large-cap tilt: the liquidity filter naturally selects the most traded, largest companies

Sector concentration: liquidity rankings often favor tech and financials.



Let’s code!

Hands-On: Your First Data Pipeline

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